Documentation for Assignment 3

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Data Set 1:

1. **Importing Libraries:**

```
        import numpy as np
        import matplotlib.pyplot as plt
```

2. **Reading Data and Processing :**

```
    with open("dataset1.txt","r") as f:
    f=f.read()
    fp=f.split()
    for i in range(0,len(fp),2):
        t.append(float(fp[i]))
        g.append(float(fp[i+1]))
    t=np.array(t)
    g=np.array(g)
```

   - The code reads data from a file named `dataset1.txt` as `f`.
   - The data is stored in two separate lists `t` and `g`.
   - The data is read and split into list.
   - The even elements are appended to `t` and the odd elements to `g`.
   - This way I sorted data into `t` and `g`.
   - Then, `t` and `g` are converted to NumPy arrays because with NumPy arrays we can perform operations like element wise multiplication.

3. **Linear Regression Function:**

   - `stline(x, m, c)` is a function that computes a straight line given `x`, `m` (slope), and `c` (intercept).
   - Now trying to solve, we should find the M matrix for this we used

```
            M = np.column_stack([t, np.ones(len(t))])
```

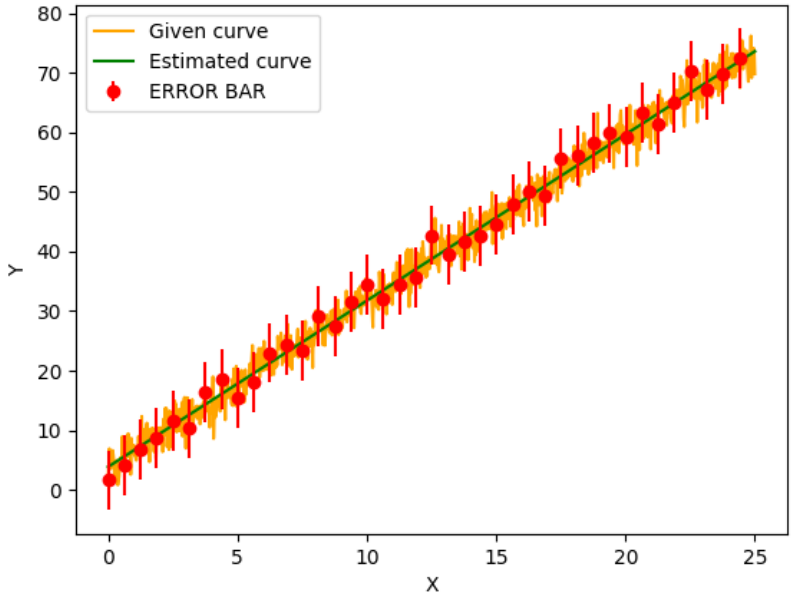
   - `columnstack()` function in NumPy is used to arrange 1-D input arrays as columns into a 2-D array by adding anothor column with ones.

- The code performs linear regression on the data to find the best-fit line equation. It uses `np.linalg.lstsq` to find the least squares solution.

4. **Results:**

- The coefficients of the best-fit line (`p1` and `p2`) are printed.

```
The estimated equation is
(2.791124245414918) t+ (3.848800101430743)
```



- The plot is saved as `data1pic.png`.

Data Set 2:

Reading and Processing the data is same as in Data Set 1 but the data is taken from `dataset2.txt`.

1. **Finding frequencies:**

   - I used plot of the Original data given in `dataset2.txt` to find the lowest frequency (Highest timeperiod) of the three sine curves.
   - I found the lowest frequency by finding the distance between two points on the line Y=0 and curve. Then their difference gives me the value of highest timeperiod.
   - Like this i found timeperiod [T=2.5007s] and frequency [fq=0.399].
   - As mentioned there are 3 sine curves with two of them with timeperiods as T and T/3. To find the timeperiod of the thrid sine curve I analysed the given plot and thought that they should be harmonic to get such a curve so, by this timeperiod of thrid sine curve is T/5.

2. **Function f1() :**

```
    def f1(x, w, p0, p1, p2):
        return p0 * np.sin(x*w) + p1 * np.sin(3*w*x) + p2
    * np.sin(5*w*x)
```

   - x is the list of first column in the given data,[p0,p1,p2] are coefficients of given in the function
   - w=(2*pi/T)=2.511296837

3. **Doing it using lstsq :**

   - To find coeff's using this method we need to make M matrix.
   - M matrix is formed by code as we did in last data set.

```
        M = np.column_stack([np.sin(x*w), np.sin(3*w*x),
    np.sin(5*w*x)])
```

   - For solving coeff's I used same function :

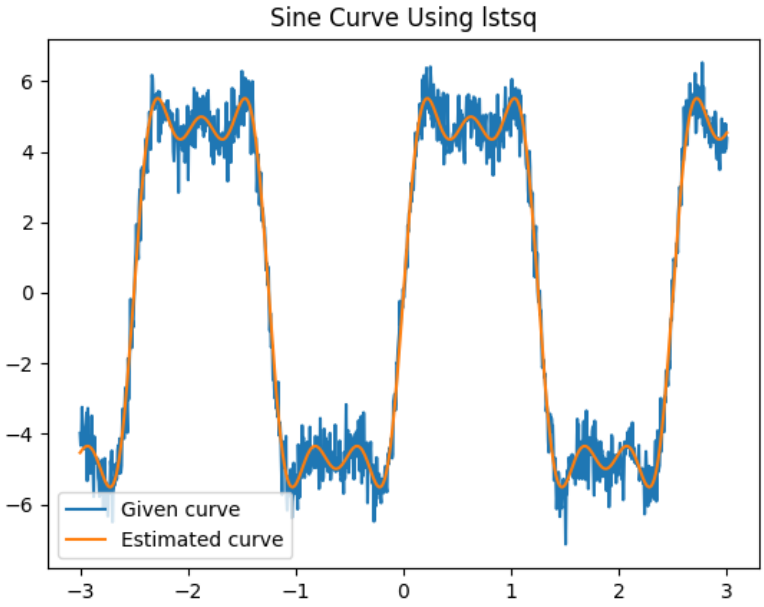
```
        (p0,p1,p2),_,_,_= np.linalg.lstsq(M, y, rcond=None
    )

```

   - This function will solve the coeff's and returns them in the first tuple.
   - The result of this method is:

```
        The estimated parameters are:
        6.01110857890662 sin(2.5125706031029655(x)) +
    2.001429758789981 sin(7.537711809308897(x)) +
    0.9811050902663911 sin(12.562853015514827(x))
```

- The plot using this lstsq method is



4. **Doing it using curvefit() :**

- After importing optimize from scify and performing the codes with assumption w1,p0,p1,p2 as 2,5,1,1.

```
        (w1, p_0, p_1, p_2), _ = optimize.curve_fit(f1, x,
    y, p0=(2,5,1,1))

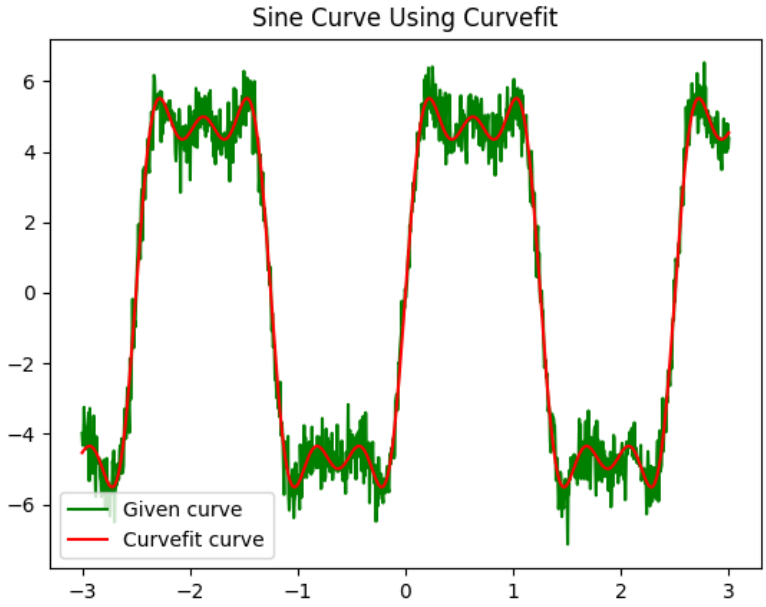
        ycurfit=f1(x,w1,p_0, p_1, p_2)
```

- After finding out the values the results are as

```
            The estimated parameters by curvefit are:
6.0111200110217355 sin(2.5127345135626733(x)) +
2.0014587685296408 sin(7.53820354068802(x)) +
0.9809066635376205 sin(12.563672567813366(x))

```

- The plot using this curvefit method is



5. **Differences :**

   - There is a small difference between the amplitude of the sine curves and also there is difference between the angular frequency.
   - Errors are very less i.e, the error in the first amplitude is Error=0.00001143211511582

Data Set 3-1:

Reading and Processing the data is same as in Data Set 1 but the data is taken from `dataset3.txt`.

1. **Function black-body() :**

```
def black_body(fq,T):
return (2 * h * (fq**3) / (c**2)) / (np.exp((h * fq) /
 (k * T)) - 1)

```

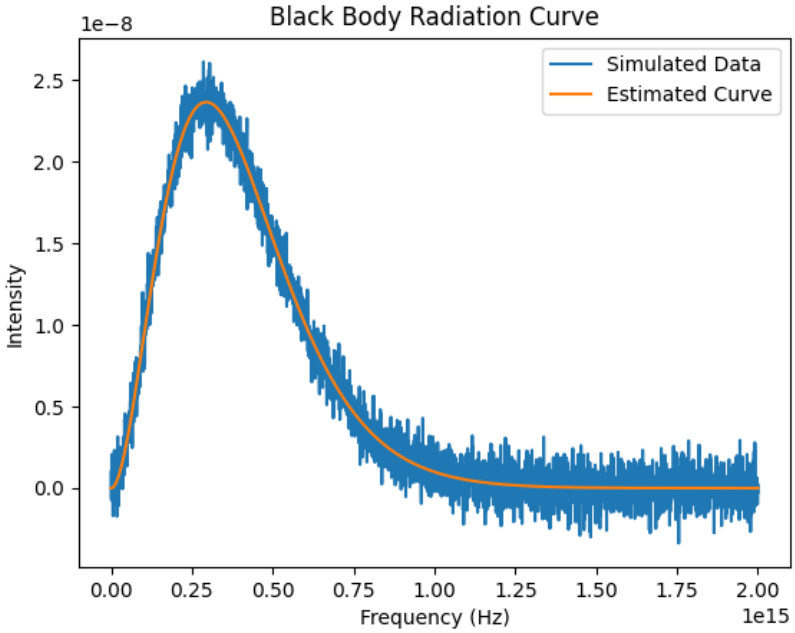
- I first took the values of h,k,c to their original values.
- This function returns the value of intensity of black body radiation.

2. **Estimating Using Curvefit :**

```
        (T),_= curve_fit(black_body, fq, I, p0=300)
```

- In the above I assumed p0=300(also if we give any resonable value we get the same result) which is the assumption of value of T.
- The function curvefit will perform operations and find the value of actual T converges the curve.
- Result is as shown below

```
        The estimated temperature is approximately
[5001.41337488] K
```



Data Set 3-2:

Reading and Processing the data is same as in Data Set 3-1.

1. **Function black-body() :**

```
def black_body(fq,T,h,k,c):
return (2 * h * (fq**3) / (c**2)) / (np.exp((h * fq) /
 (k * T)) - 1)
```

   - In the above function this time we take T,h,k,c as variables.

2. **Estimating Using Curvefit :**

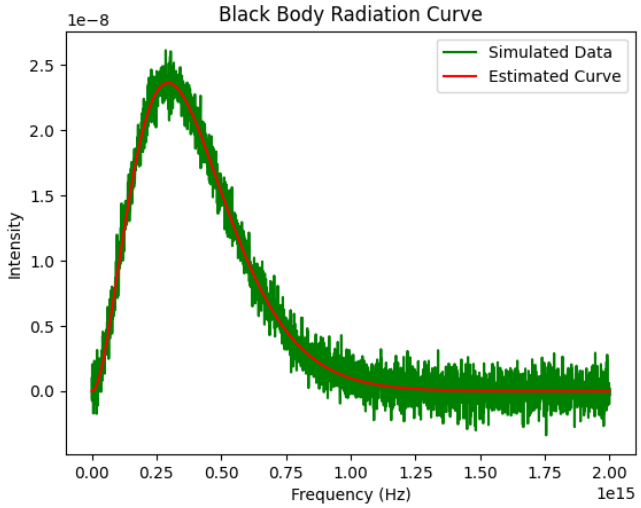
```
(T,h,k,c),_= curve_fit(black_body, fq, I, p0
=(4549,6.45e-34,1.3e-23,2.45e8)
```

   - This time we give curvefit four set of values as T=4549, h=6.45e-34, k=1.3e-23 and c=2.45e8
   - I chose this values because we after testing so many sets I am getting most converging values for this set.
   - The result of this set of values with curvefit is

```
    The estimated temperature is approximately
5001.4213997662455 K
    The estimated Plank's constant is
6.924782436068668e-34,
    The estimated Blotzmann's constant is
1.4430585100392132e-23,
    The estimated speed of light is 307037573.9978961

```

   - The Resulting curve for this is :



3. **Quality of Results :**

- To get the Given values I have tried so many sets and this the best of the resluts I got.
- The results keep on changing as we change any small value of assumption set so it is very important to choose reasonably close starting points for our estimates in order to make it converge.
- I got both the temperatures almost equal and with very small error.[Error=0.00802488624]
- The actual value of plank's constant is 6.626e-34 but I got 6.924e-34. Here also error is quite small it is 0.298e-34.
- The actual value of Blotzmann's constant is 1.38e-23 but I got 1.44e-23.Here the error is 0.06e-23.
- The actual value of c is 3e8 but I got 3.07e8.The Error is 0.07e8.